

Arithmetic Operators in Java

Arithmetic operators are used to perform basic mathematical operations like addition, subtraction, multiplication, division, and modulus. They operate on numerical values (primitives like `int`, `float`, etc.).

1. List of Arithmetic Operators

Operator	Description	Example
<code>+</code>	Addition	<code>a + b</code>
<code>-</code>	Subtraction	<code>a - b</code>
<code>*</code>	Multiplication	<code>a * b</code>
<code>/</code>	Division	<code>a / b</code>
<code>%</code>	Modulus (remainder)	<code>a % b</code>

2. Description of Operators

1. Addition (`+`)

- Adds two numbers.
- Can also concatenate strings.

Example:

```
public class AdditionExample {  
    public static void main(String[] args) {  
        int a = 10, b = 20;  
        int sum = a + b;  
        System.out.println("Sum: " + sum);  
    }  
}
```

2. Subtraction (`-`)

- Subtracts one number from another.

Example:

```
public class SubtractionExample {  
    public static void main(String[] args) {  
        int a = 20, b = 10;  
        int difference = a - b;  
        System.out.println("Difference: " + difference);  
    }  
}
```

3. Multiplication (*)

- Multiplies two numbers.

Example:

```
public class MultiplicationExample {  
    public static void main(String[] args) {  
        int a = 5, b = 4;  
        int product = a * b;  
        System.out.println("Product: " + product);  
    }  
}
```

4. Division (/)

- Divides one number by another.
- Produces a quotient.
- For integers, fractional parts are truncated (e.g., $5 / 2 = 2$).

Example:

```
public class DivisionExample {  
    public static void main(String[] args) {  
        int a = 10, b = 3;  
        int quotient = a / b; // Integer division  
        System.out.println("Quotient: " + quotient);  
  
        double x = 10.0, y = 3.0;
```

```
        double result = x / y; // Floating-point division
        System.out.println("Result: " + result);
    }
}
```

5. Modulus (%)

- Returns the remainder of a division operation.
- Commonly used for checking divisibility or extracting digits.

Example:

```
public class ModulusExample {
    public static void main(String[] args) {
        int a = 10, b = 3;
        int remainder = a % b;
        System.out.println("Remainder: " + remainder);
    }
}
```

3. Key Points

1. Order of Operations (Precedence):

- Multiplication (*), Division (/), and Modulus (%) have higher precedence than Addition (+) and Subtraction (-).
- Operators with the same precedence are evaluated from **left to right**.

Example:

```
public class OperatorPrecedence {
    public static void main(String[] args) {
        int result = 10 + 20 * 3 / 2 - 5;
        System.out.println("Result: " + result); // Evaluates to 35
    }
}
```

2. Type Compatibility:

- When performing arithmetic operations on different data types, smaller types (byte, short) are promoted to int.

- For operations involving `float` or `double`, the result is promoted to the higher precision type.

4. Comprehensive Example

```
public class ArithmeticOperatorsDemo {  
    public static void main(String[] args) {  
        int a = 15, b = 4;  
        System.out.println("Addition: " + (a + b));           // 19  
        System.out.println("Subtraction: " + (a - b));        // 11  
        System.out.println("Multiplication: " + (a * b));      // 60  
        System.out.println("Division: " + (a / b));            // 3  
        System.out.println("Modulus: " + (a % b));             // 3  
    }  
}
```

Output:

```
Addition: 19  
Subtraction: 11  
Multiplication: 60  
Division: 3  
Modulus: 3
```